Study Guide: Stepwise Selection & Regularization (ISLR Chapter-6)

Expect a random sampling of questions of the following types (not every question type will be represented)

Question: You should expect questions asking you to evaluate whether different model selection methods are practical for large numbers of features and whether they can be used when the number of features exceeds the number of observations ((p>n)). Typical questions will ask you to decide, for each method, whether it is computationally feasible and/or whether it actually can work when (p>n).

You should understand the following:

- Best subset selection evaluates all (2^p) possible models. It becomes infeasible for large (p), but it can still work when (p>n) because it considers small subsets.
- Forward stepwise selection adds variables one at a time. It is much faster than best subset and works when (p>n).
- Backward stepwise selection removes variables starting from the full model. It requires fitting the full model first, so it cannot be used when (p>n).

Question: You should expect questions comparing best subset, forward selection, and backward selection.

These questions may ask you which method is most appropriate in different settings and why.

You should be able to explain:

- How each method searches the model space
- How many models it evaluates
- What assumptions it makes
- What its main limitations are

You may be asked to justify your answer using computational or statistical reasoning.

Question: You should expect questions where you compute norms of coefficient (parameters) vectors.

These will usually involve calculating (L^1) and (L^2) norms from given parameter values.

You should know:

$$|\beta|_1 = \sum_j |\beta_j| \tag{1}$$

$$|\beta|_2 = \sqrt{\sum_j \beta_j^2} \tag{2}$$

Ridge regression uses ($|\beta|_2^2$), lasso uses ($|\beta|_1$),

You should be comfortable doing these calculations quickly and accurately.

Question: You should expect questions comparing two models under regularization when their un-regularized loss is the same.

These questions will ask which model is preferred under Ridge or Lasso for a fixed ($\lambda > 0$).

You should understand that:

- Ridge minimizes: loss + $(\lambda|\beta|_2^2)$
- Lasso minimizes: loss + $(\lambda|\beta|_1)$

When losses are equal, the model with the smaller penalty term is preferred (since it minimizes the total objective, i.e. the sum of both terms).

Question: You should expect conceptual questions about how Ridge and Lasso affect coefficients.

These questions may ask which method is more likely to set coefficients to zero and why.

You should know:

- Ridge shrinks coefficients smoothly and rarely produces exact zeros.
- Lasso encourages sparsity and often sets some coefficients to exactly zero.
- Lasso performs automatic feature selection, while Ridge does not.

You should be able to explain this in words, not just select an answer.

Question: You should expect questions involving the geometric interpretation of regularization in low dimensional parameter spaces (e.g from matching plots in parameter space to).

These questions usually show plots of unit balls for different norms and ask which one promotes sparsity.

You should understand that regularization can be viewed as minimizing loss subject to a constraint

Different norms produce different shapes:

- (L^2): circle, smooth boundary, little sparsity
- (L^1): diamond, sharp corners on axes, strong sparsity
- (L^∞): square, weak sparsity

- (L^p) with (p<1): star-like shape, very strong sparsity

Sparsity occurs because solutions tend to land on corners aligned with coordinate axes.

Question: You should expect questions asking which penalties produce sparse models and which produce dense models.

These questions may ask you to rank norms by how aggressively they remove features.

You should know:

- (L^2): dense models, keeps most variables
- (L^1): sparse models, convex optimization
- (L^p) for (p<1): very sparse, non-convex optimization

Question: Forward Stepwise Selection: Iteration Questions

Expect a question where you need to:

- Perform the first few steps of forward stepwise selection using provided numerical values.
- Compare models using a criterion such as RSS, MSE, or validation error.
- Select variables step by step based on which model performs best.
- Keep track of which predictors are allowed to enter the model at each stage.

Typical Question Format:

You will be given:

- A small set of predictors (for example, X_1, X_2, X_3).
- A table showing model performance for different combinations of variables.
- A rule such as "choose the model with the smallest RSS."

You will be asked to carry out one or two iterations of forward selection.

What You Should Be Able to Do:

1. First iteration

- Compare all single-predictor models.
- Identify which predictor gives the best performance.
- Select that variable.

2. Second iteration

- Consider only models that include the selected variable.
- Compare the remaining candidate models.

- Choose the variable that leads to the best improvement.

3. Report the current model after two steps.

Best Subset Selection: Iteration Questions

Expect a question where you need to:

- Perform model comparison using best subset selection.
- Compare models of the same size (same number of predictors).
- Identify the best model at each subset size.
- Be able to distinguish clearly between best subset and forward selection.

Typical Question Format:

You will be given:

- A small set of predictors (e.g., X_1, X_2, X_3).
- A table of RSS (or MSE) values for all possible subsets.
- A rule such as "choose the model with the smallest RSS."

You will be asked to:

- Identify the best 1 -variable model.
- Identify the best 2 -variable model.
- Possibly compare across model sizes.

Question: Model Selection Criteria: Conceptual Understanding

Expect a question where you need to:

Explain what AIC, BIC, Mallows' C_p, and adjusted R^2 measure, how they are used for model selection, and how they balance model fit and complexity.

Typical Question Format: You may be asked something like:

For each of the following model selection criteria-AIC, BIC, C_p, and adjusted R^2-briefly describe:

- What quantity it measures or approximates
- Whether smaller or larger values are preferred, and
- How it penalizes model complexity.
- What it is based on (likelihood, RSS, etc),
- What type of model it tends to favor (simpler vs. more complex).
- How its formula discourages overly complex models,